

The Hodgkin-Huxley model

Conductance models of the action potential

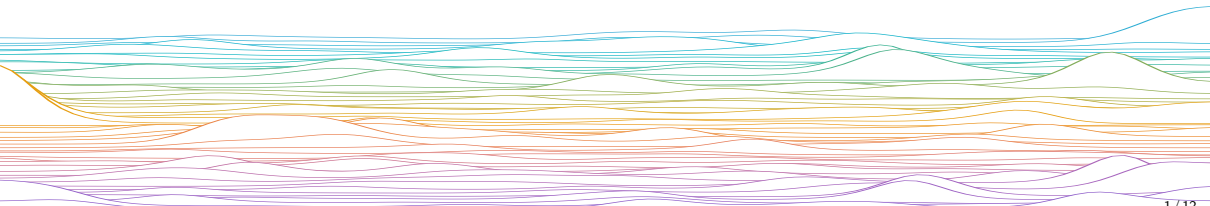
Computational Neuroscience

University of Bristol

M Rule

Learning outcomes:

- Understand and describe a basic voltage clamp setup.
- Be ready to approach exam problems that touch on the voltage clamp.



Hodgkin and Huxley fit them to match their voltage-clamp data.

$$\begin{aligned}\alpha_m(V) &= \frac{0.1(V + 40)}{1 - e^{-(V+40)/10}} & \beta_m(V) &= 4e^{-(V+65)/18} \\ \alpha_h(V) &= 0.07e^{-(V+65)/20} & \beta_h(V) &= \frac{1}{1 + e^{-(V+35)/10}} \\ \alpha_n(V) &= \frac{0.01(V + 55)}{1 - e^{-(V+55)/10}} & \beta_n(V) &= 0.125e^{-(V+65)/80}\end{aligned}$$

Neuroscientists say “*clamped*”

- ▶ Voltage clamp
- ▶ Current clamp
- ▶ Dynamic clamp
- ▶ Patch clamp
- ▶ Space clamp

What do they mean by that?

Clamp :

- ▶ ‘*We fixed some variables at known values, so that the system is easier to analyze or identify*’
- ▶ Experimentally:
 - Using feedback control
- ▶ Mathematically:
 - Making assumptions

Voltage Clamp

- ▶ Get an electrode connected to the inside of a neuron (somehow?)
- ▶ Use a feedback amplifier to hold membrane at some target v_{ref}
- ▶ What can we measure?
 - Voltage-gated conductances stabilize at a steady state

Intracellular recording

Sharp Electrode

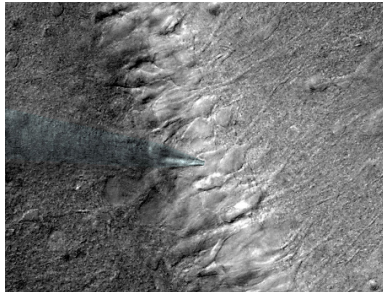
Cell Attached

Whole-Cell

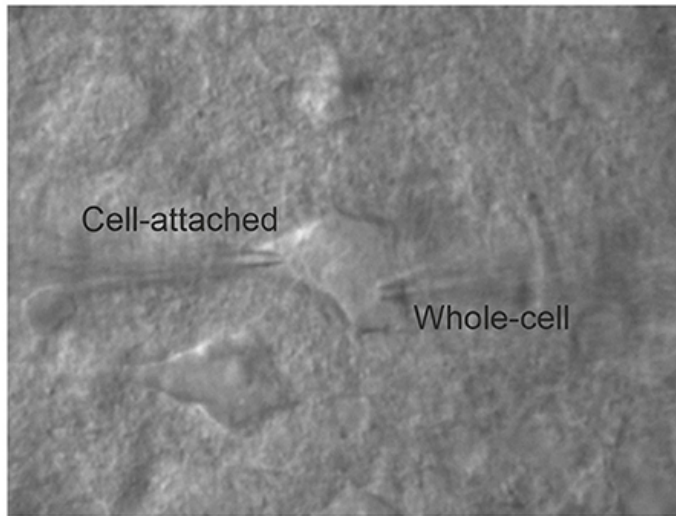
Common misunderstanding: **Patch clamp** refers to a cell-attached recording, combined with the use of a feedback amplifier to “clamp” either the voltage or current (or something more exotic) to a known value to study membrane, channel properties. *Patch clamp* $\neq$ *sharp electrode, whole cell*

In vitro electrophysiology: patch clamp

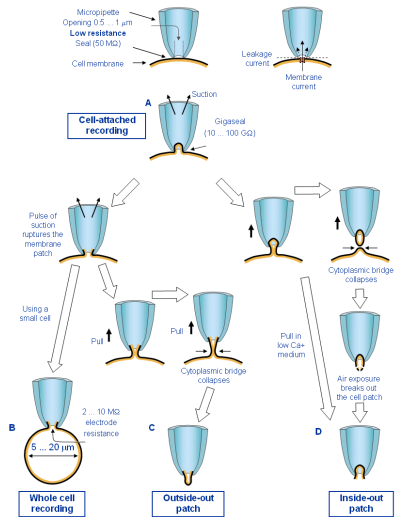
- ▶ Extracellular electrophysiology can detect action potentials. But it cannot see 'subthreshold' voltage signals like PSPs.
- ▶ To record intracellular voltage researcher use a method called the patch clamp.
- ▶ Involves sealing a cylindrical glass pipette (with an electrode inside) onto the neuron's membrane and bursting a small hole through to allow the electrode to measure the intracellular voltage.
- ▶ If the seal is not physically stable it will break. So the method is difficult to do in vivo.
- ▶ Most often done in *in vitro* brain slices or in neurons grown in a petri dish.



Bonni et al., modified from Rosentod, [via wikipedia](#)



Vazetdinova, Alina, et al. (2002)



Jaakko and Plonsey (1995)

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